

A top-down view of a grey desk with various medical and study items: a stethoscope, a laptop displaying a medical website, a tablet, a pair of glasses, a smartphone, a spiral notebook with a pen, and a coffee cup. A large white pill-shaped graphic is centered over the laptop.

Pharm*Achieve*

PAIN MANAGEMENT

Qualifying Exam Preparatory Course

COMPETENCIES COVERED IN THIS PRESENTATION...

**Providing Care:
Clinical Care**

**Providing Care:
Distribution**

**Knowledge
& Expertise**

**Communication
& Collaboration**

**Leadership &
Stewardship**

Professionalism

LEGEND

5-HT Serotonin

ADRs Adverse Drug Reactions

ASA Acetylsalicylic Acid

CNS Central Nervous System

COX Cyclooxygenase

CYP Cytochrome P450

GI Gastrointestinal

IM Intramuscular

IR Immediate Release

IV Intravenous

LA Long-Acting

NE Norepinephrine

NeP Neuropathic Pain

NMDA N-Methyl-D-Aspartate

NSAID Nonsteroidal Anti-inflammatory Drug

N/V Nausea/Vomiting

OA Osteoarthritis

OIC Opioid-Induced Constipation

PG Prostaglandin

PO By Mouth

PRN As Needed

Q Every

SC Subcutaneous

SL Sublingual

SNRI Serotonin-Norepinephrine Reuptake Inhibitor

SNS Sympathetic Nervous System

SSRI Selective Serotonin Reuptake Inhibitor

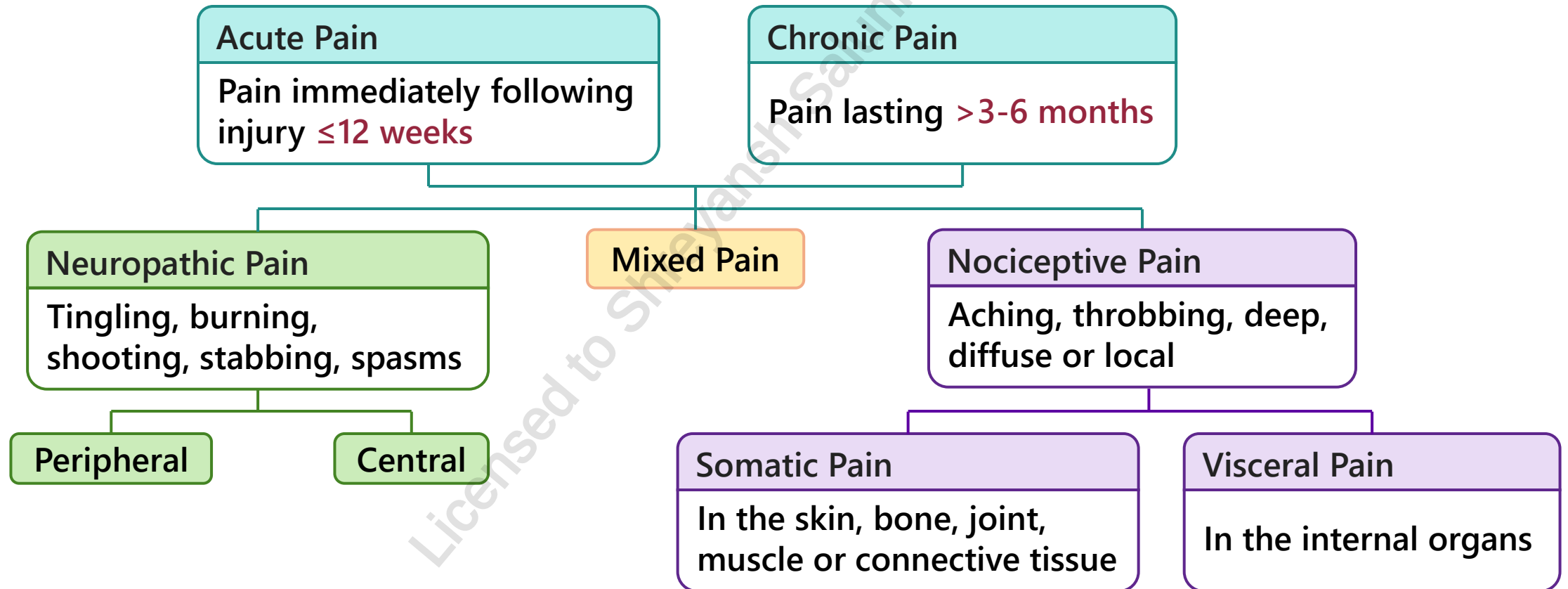
TCA Tricyclic Antidepressant

DEFINITIONS

- **Allodynia:** painful response to non-noxious stimuli
- **Hyperalgesia:** increased pain response to noxious stimuli
- **Incident pain:** pain resulting from movement/activity
- **Tolerance:** diminished analgesic effect over time
- **Breakthrough pain:** pain eruption during maintenance analgesic therapy
- **Physical dependence:** withdrawal symptoms from discontinuation of tolerance causing therapy

PAIN CLASSIFICATION ON DURATION & CHARACTERISTICS

Possible symptoms of **untreated pain**:
tachycardia, diaphoresis, HTN, pallor, ↑SNS activity, ↓range of motion



CAUSES OF PAIN

PAIN CAUSES

- Incident
 - Diseases such as migraine disorders, multiple sclerosis, trigeminal neuralgia, and many others
 - Injuries that result from things such as falls, trauma, car accidents, etc.
- Medications such as chemotherapy agents, statins, bisphosphonates, etc.
- Mood disorders

GOALS OF THERAPY

- ✓ Treat underlying cause of pain
- ✓ Manage pain and improve function using a multi-modal approach
- ✓ Minimize risk of ADRs
- ✓ Manage concurrent conditions; insomnia, mood disorders

NON-PHARMACOLOGICAL MEASURES

Behavioural and Psychological Modalities

- Pain education
- Cognitive behavioural therapy (CBT)
- Mindfulness-Based Stress Reduction (MBSR)
- Relaxation and stress management techniques

Lifestyle Interventions

- Sleep hygiene
- Healthy diet
- Work-life balance

Physical Modalities

- Exercise
- Physiotherapy
- Ice or heat therapy
- TENS
- Massage
- Acupuncture

TOPICAL AND LOCALIZED THERAPIES

Capsaicin (topical)

- Initial burning sensation that dissipates over time but then attenuates cutaneous hypersensitivity (local blistering, peeling, or redness of skin)

Diclofenac 1.16%, 2.32% (topical NSAID for MSK pain)

- Local anti-inflammatory
- Safer than oral NSAID, decreased incidence of GI effects

Lidocaine (local injections)

- Numbing agent

Steroids (local injections)

- Anti-inflammatory

PHARMACOLOGICAL OVERVIEW

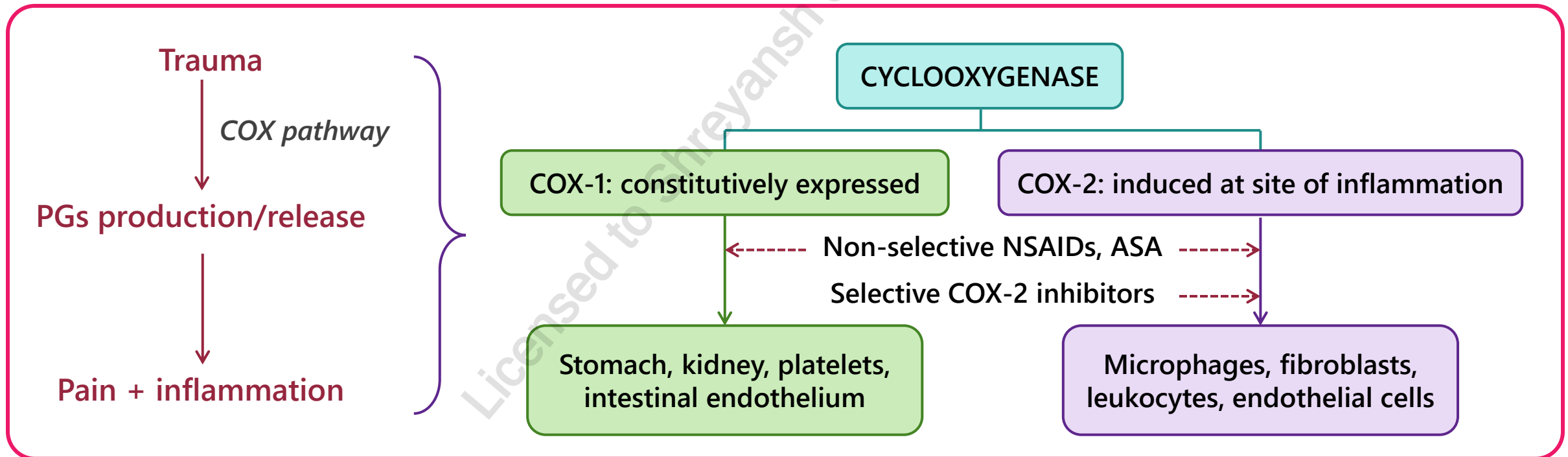
Drug	Place in Therapy	
Acetaminophen	• 1st line: mild pain, post-operative pain • No effect in NeP	Nociceptive pain
NSAIDs	• 1st line: OA, mild-moderate nociceptive pain/acute inflammation • No effect in NeP	
TCA	• 1st line: many limiting side effects	Neuropathic pain
Gabapentin Pregabalin	• 1st line: diabetic peripheral neuropathy, postherpetic neuralgia, spinal injuries	
Duloxetine	• 1st line: diabetic peripheral neuropathy, fibromyalgia, chronic low back pain	
Opioids	• 2nd-3rd line: severe nociceptive pain, adjunct therapy	

ACETAMINOPHEN SUMMARY

- Indication → **Nociceptive pain** [No effect in NeP]
- **1st line for mild pain and post-operative pain**
 - No peripheral anti-inflammatory activity (only central prostaglandin inhibition)
- **Ceiling effect → 1000 mg/per dose**
- **Maximum dose → 4000 mg/day**
 - Caution with concomitant use of cough/cold products containing acetaminophen
 - Counsel patients to track their daily acetaminophen levels when using these products
- Risk of hepatotoxicity → **max 2000-3000 mg/day**
 - Caution in older adults, hepatic impairment, history of alcohol abuse, overdose
- Warfarin patients
 - Chronic use of acetaminophen **doses 1.3-2 g/day** can enhance the anticoagulation effect of warfarin (*Risk C: monitor therapy*)

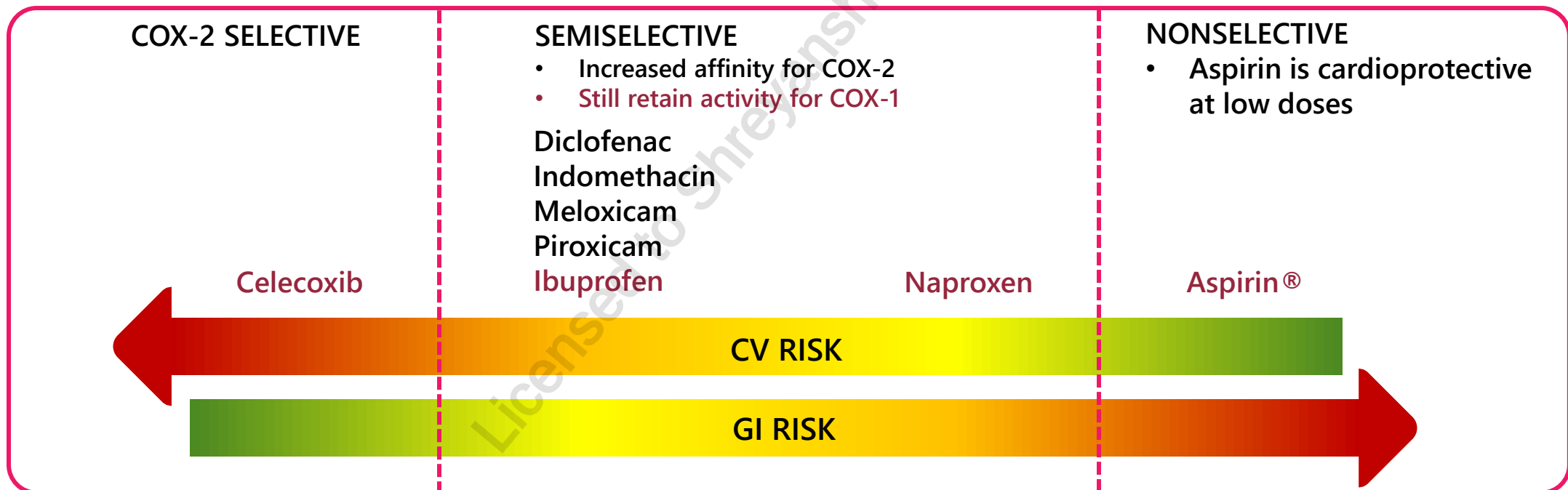
NSAID SUMMARY & PHARMCODYNAMICS

- Indication -> **Nociceptive pain** [No effect in NeP]
- **1st line for mild-moderate pain, OA, acute inflammation, headaches, and menstrual pain**
- **Prescribe cautiously in elderly and patients with cardiac, renal, and GI risk factors**
 - Examples include patients on anticoagulants, patients with chronic kidney disease, and patients with previous GI bleeding
- **Avoid** in 3rd trimester of pregnancy (including topical NSAIDs)



NSAID SELECTION

- For patients with a high risk of GI side effects (e.g. history of ulcer or UGIB, smoking, alcohol), using a COX-2 inhibitor along with a proton pump inhibitor (PPI) should be considered
- If patients have both cardiac and GI risks factors, weigh risk versus benefit of using NSAID:
 - If **GI risk** is the primary concern: **Celecoxib + PPI**
 - If **CV risk** is the primary concern: **Naproxen + PPI**



OPIOID PHARMACOLOGICAL SUMMARY

Medication	Clinical Pearls
Morphine	Gold standard opioid to which others are compared Avoid in advanced CKD (CrCl <60 mL/min) due to risk of metabolite accumulation
Codeine	Least potent opioid (200mg codeine=30mg morphine), prodrug that requires CYP2D6 metabolism to activate to morphine (genetic polymorphisms impact effect of therapy!) Avoid in breastfeeding and close to term, avoid in advanced CKD
Tramadol	Multimodal effect at 5-HT and NE receptors, effective for neuropathic pain Potency is less than morphine, CYP2D6 mediated , 90% excreted in urine
Hydromorphone*	May have better tolerability (5X more potent than morphine) • First-line/preferred opioid in renally impaired patients
Oxycodone*	Increased risk of misuse (2X more potent than morphine). Dose adjust in CKD
Fentanyl*	Transdermal patch applied every three days , takes ~ 12-24 hours for onset with patch Potent, inter-patient variability for absorption, not for use in opioid naïve patients
Buprenorphine*	Lipophilic, partial opioid agonist May be used in opioid naïve patients for moderate persistent pain, SL tablet and transdermal patch

*can be used in patients with CKD



Evaluate risk versus benefit of opioid use during pregnancy – shortest duration and avoid close to delivery d/t risk of neonatal respiratory depression

OPIOID PHARMACOLOGICAL SUMMARY

Medication	Key Points
Hydrocodone	High doses required in analgesia, commonly used for cough suppression
Heroin	Highly addictive and may be used illicitly
Meperidine	Has a toxic metabolite that may accumulate in renal dysfunction (seizure risk) , so should be avoided for pain control in these patients
Methadone*	Long half-life which made lead to accumulation; therefore, only titrate q3-7 days Primarily used for opioid use disorder and palliative pain control Very potent, many drug interactions
Pentazocine (mixed agonist antagonist)	Low efficacy, mixed agonist/antagonist May precipitate withdrawal or reduced analgesic efficacy if taking pure opioid agonists

*can be used in patients with CKD

OPIOID MANAGEMENT

There is no maximum dose. Titration is required,

1. Start with short acting agent + PRN to determine total daily dose
2. Then initiate LA agent once appropriate dose is determined

Short-Acting Agent

- Use **lowest dose** and then adjust dose to patient pain control
- Examples: Oxycodone IR, Hydromorphone IR, Morphine IR, Oxycocet®, Lenoltec #3
- Give throughout 24-hour period

Long-Acting Agent

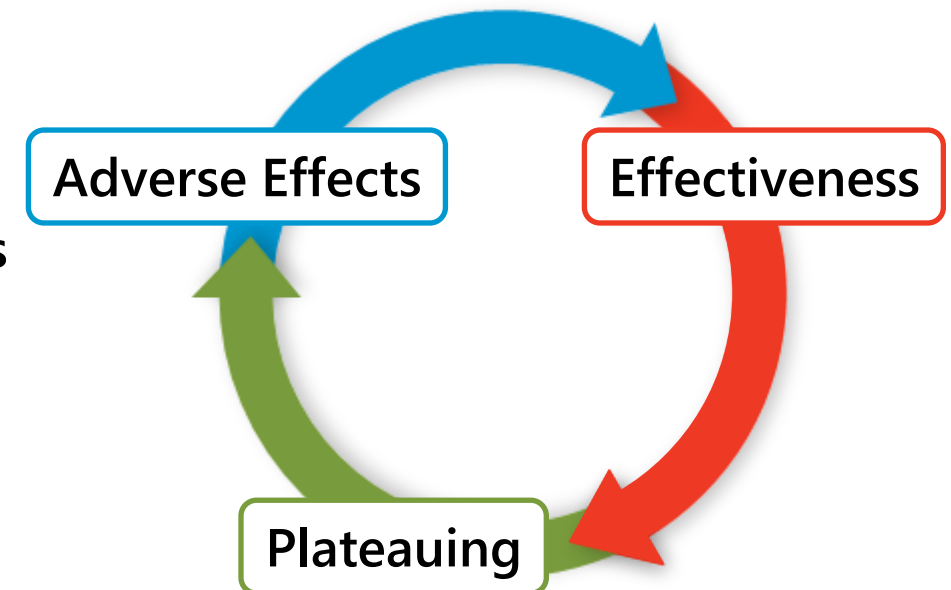
- Dose based on 24-hour use of fast-acting agents
- **Provides foundational pain management**
- Reassess maintenance dose **and** **↑ dose according to breakthrough use**

Breakthrough Pain

- Use **PRN dosing, immediate-release formulations with short $T_{1/2}$**
- Breakthrough pain opioid dose = **10-20% total daily dose given q1-2H SC/IV or q3-4H PO PRN**
- **↑ scheduled doses if using breakthrough medication 4-6x/day**

OPIOID OPTIMAL DOSE

- Start with low dose **then increase dose gradually while monitoring** effectiveness until optimal dose reached
 - End titration when:
 - Optimal dose is reached OR
 - Patient reports insufficient pain relief after 2-3 dose increases OR
 - Intolerable side effects OR
 - Misuse/addiction
- **Goal: Pain reduction of $\geq 30\%$**
 - Important to set realistic expectations with patients
 - To find optimal dose, need to balance:



OPIOIDS COMPLICATIONS

Opioid Complication	Description
Worsens sleep apnea	Obstructive and more often central sleep apnea
Opioid-induced hyperalgesia (OIH)	Increased sensitivity to pain stimuli
Neuroendocrine abnormalities	Secondary hypogonadism that could result in sexual dysfunction or infertility, mood disorders, osteoporosis
Respiratory depression	Characterized by reduced respiratory rate
CNS depression	Sedation, lack of coordination, delirium, coma
Opioid-induced constipation (OIC)	Inhibition of gastric emptying and peristalsis in GI tract

MANAGING OPIOID SIDE EFFECT MANAGEMENT

Side Effects	Management
Constipation	Hydration, exercise, stimulant laxative (senna/bisacodyl), osmotic agent (PEG 3350, lactulose) If refractory OIC, may use methylnaltrexone Tolerance does not develop
Nausea/ vomiting	Dimenhydrinate, ondansetron, haloperidol, domperidone, metoclopramide (tolerance occurs quickly and persistent nausea is not common) If conventional agent not tolerated, olanzapine or risperidone may be considered
Sedation & Neurotoxicity	Stop unnecessary CNS depressants and hold or reduce opioid dose depending on severity Sedation tolerance usually develops within days to weeks Try low-dose benzodiazepines if myoclonus (brief involuntary muscle contractions)
Respiratory depression	Hold opioid Should not normally occur if you are slowly titrating up the dose for pain control Can quickly be reversed with naloxone (opioid antagonist)
Withdrawal*	Taper: Multiple approaches; OTC agents, clonidine (autonomic symptoms)

*Withdrawal symptoms: FLOP [flu-like symptoms, lethargy, ongoing pain, panic (anxiety/agitation, sweats, insomnia)]

- Occurs 6-36h after last dose; ensure to counsel when switching opioids

OPIOID DRUG INTERACTIONS

INTERACTING DRUG	ADVERSE EFFECT
Serotonergic Drugs (e.g., SSRI, SNRI, TCAs, phenelzine, MAO inhibitors, Triptans)	Serotonin syndrome MAO-I contraindicated within 14 days of opioid usage
Respiratory depressants (e.g. benzodiazepines, alcohol, barbiturates, opioids)	Slows the brain's respiratory drive leading to increased respiratory depression (shallow breathing, decreased respiratory rate)
CNS depressants (e.g., antidepressants, antipsychotics, benzodiazepines, alcohol gabapentinoids)	Sedation Falls Cognitive impairment

SWITCHING OPIOID THERAPY

When should opioid therapy be switched?

- 1. Patient has **dose-related toxicity**
- 2. High **likelihood of poor efficacy**
- 3. Specific **formulation** that is not available for the current drug
- 4. Room for **significant dose reduction**
- 5. Ineffective for pain management
- 6. Intolerable **SE**

How is opioid therapy switched?

MORPHINE EQUIVALENCE	
Drug	Oral Dose (mg)
Morphine	30
Oxycodone	20
Hydromorphone	6
Codeine	200
Tapentadol	100
Tramadol and Methadone	Morphine dose equivalence not reliably established

INCOMPLETE CROSS-TOLERANCE

Incomplete cross-tolerance

Generally, after long-term exposure to a medication, patients become cross-tolerant to structurally similar medications; however, this is rarely a complete effect with opioids. Therefore, once you have finished calculating a morphine equivalence:

Dose must be ↓ by 25-50%* when switching to a different opioid (such as hydrocodone to oxycodone, etc.)

Exceptions

- Methadone: morphine equivalence is not reliably established
- Fentanyl: the conversion chart already accounts for incomplete cross-tolerance

This does NOT apply when switching from IV/IM/SC to PO of the SAME molecule!
Example: a patient discharged from the hospital on PO morphine would not require a dose reduction from the IV morphine they received in the hospital

FENTANYL CONVERSION

Morphine PO (mg per day)	Fentanyl patch (mcg/h)
60-134	25
135-179	37 (25+12)
180-224	50
225-269	62 (50+12)
270-314	75
315-359	87 (75+12)
360-404	100

UNIDIRECTIONAL

Patch Strength	Approximate Single Breakthrough dosing: Morphine IR
12 mcg/hr	5 mg
25 mcg/hr	10 mg
50 mcg/hr	20 mg
75 mcg/hr	25 mg
100 mcg/hr	35 mg



Patch is contraindicated in post-operative, acute pain, and opioid-naïve patients



Chart above already accounts for cross-tolerance!
So no need for further dose reduction by 25-50%

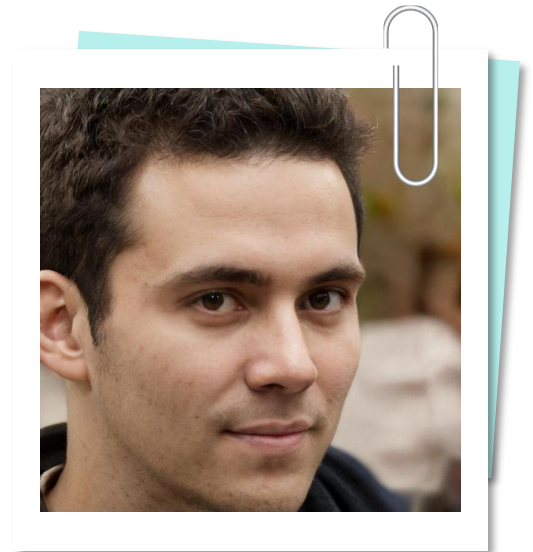
PRACTICE QUESTION 1

SA has been using morphine SR 60mg q12h and morphine IR 10mg q4hprn (approximately 6 tablets a day). He has been complaining of poor pain control and ongoing nausea from the morphine.

Dr. Smith wants SA to switch to hydromorphone CR and asks you for a dose recommendation.

What would you recommend?

- a) Hydromorphone contin 18mg q12h
- b) Hydromorphone contin 9mg q12h
- c) Hydromorphone contin 3mg q12h
- d) Hydromorphone contin 24mg q12h



PRACTICE QUESTION 2

LM is a 52-year-old female being treated for chronic pain. Her current medications include oxycodone CR 80mg twice daily and oxycodone 10mg every 4-6 hours as needed for breakthrough pain. LM states she usually requires 4 doses of her breakthrough medication. Her doctor would like to switch her to a fentanyl patch due to her recent difficulties with swallowing.

1. Which of the following is an appropriate dose for LM's starting fentanyl patch?

- a) 25 mcg
- b) 50 mcg
- c) 75 mcg
- d) 100 mcg

2. What dose of morphine should be used for LM's breakthrough pain?

- a) 12.5 mg
- b) 20 mg
- c) 25 mg
- d) 30 mg

PRACTICE QUESTION 3

RB, 45-year-old male, was in a car accident 2 months ago when a driver rear-ended his car. He reports myofascial back and neck pain. He is currently doing physiotherapy and swimming and has found some benefits with these modalities. He started on morphine IR initially and then was stabilized on morphine SR one month ago. For the past 1 week he has noticed that this regimen wasn't as effective. He was finding it harder to engage in physiotherapy and going to work has been more challenging. He feels as though this regimen is now barely decreasing his pain from 6/10 to 4-5/10. He is tolerating the morphine well, you see no early refills, and no aberrant drug related behaviors.

His current medications include:

- Morphine SR 10mg q12h
- Celecoxib 100mg BID
- Escitalopram 10mg daily
- Pantoprazole 20mg daily

What would you recommend as the next step for his opioid management?

- a) Add Lenoltec No. 3 (1-2 tablets q4h prn)
- b) Increase morphine SR dose
- c) Switch to oxycodone CR
- d) Continue on the same dose of morphine SR



PRACTICE QUESTION 4

TR is a 68-year-old with a long-standing history of diabetes. She has a new diagnosis of diabetic neuropathy, reporting that her legs and the bottom of her feet are burning. She has trouble sleeping due to her pain. Her family doctor is looking for a good medication to start her on for peripheral neuropathy. Medical history include significant constipation, glaucoma, and tachycardia.

Which agent is the most appropriate for her to try?

- a) Nortriptyline
- b) Celecoxib
- c) Pregabalin
- d) Lenoltec No. 3



MONITORING PARAMETERS

Efficacy Endpoints		
Parameter	Degree of Change	Timeframe
Acute Pain	Improved	Improved within 1-2 hours
Chronic Pain	At least 30% reduction in pain intensity Improved function	Duration of therapy

Safety Endpoints		
Parameter	Degree of Change	Timeframe
ADR of analgesic (N/V, constipation, urinary retention, respiratory rate, drowsiness)	Prevent or minimize	Duration of therapy
Opioid misuse (aberrant drug related behavior)	Prevent or minimize	Duration of therapy
Signs of overdose/withdrawal	Prevent	3 days after initiation, then ongoing

PAIN MANAGEMENT SUMMARY

Two types of pain: **nociceptive** (muscle, internal organs, etc.), **neuropathic** (nerve)

1. Nociceptive pain: treat with **acetaminophen or NSAIDs** (select specific NSAID based on patient-specific factors), **opioids** (2nd/3rd line)
2. Neuropathic pain: treat with **pregabalin, gabapentin, duloxetine, opioids** (2nd/3rd line)

Switching between opioids is indicated if: dose-related toxicity, likelihood of poor efficacy (i.e. CYP2D6 polymorphism and codeine), specific formulation required, to allow for dose reduction (more potent opioid), ineffective or intolerable side effects

*** Remember: reduce the dose by 25-50% when switching between opioids to account for cross-tolerance**

PAIN MANAGEMENT DRUG RECAP

Tramadol

- Provides **similar analgesia compared to NSAIDs** post-operatively.
- Useful in **patients unable to tolerate other stronger opioid therapy or if NSAIDs are contraindicated**
- **Dose adjust for severe hepatic and renal disease** (90% of metabolite excreted in urine)
- **Seizure risk** (increased risk with higher doses, concurrent SSRI/SNRI/TCA/drugs that impair tramadol metabolism, history of seizure disorder, alcohol withdrawal)
- Risk of serotonin syndrome (increased risk if taking concurrent SNRI/TCA/MAOi/triptan/drugs that impair tramadol metabolism, age >65, **CYP2D6 poor metabolizer**)

NSAIDs

- **No dependence/tolerance** and helps reduce opioid dose requirements
- If one NSAID ineffective, can switch to alternative NSAID
- Effective for **“bony” pain, nociceptive and inflammatory pain**
- **Dose ceiling effect**
- Effective to use in **combination with opioids, acetaminophen for post-surgical pain**

PAIN MANAGEMENT DRUG RECAP

Duloxetine

- Onset within 1 week, trial 6-8 weeks for efficacy
- **Gradually taper (i.e., 2-4 weeks) before discontinuing** due to risk of irritability, insomnia, anxiety, headache, dizziness, paresthesia, agitation, confusion

Gabapentin/Pregabalin

- Common neurological side effects
- Particularly **useful if co-existing sleep and anxiety**
- **Trial 6-8 weeks**
- **Pregabalin has linear kinetics, better bioavailability, faster onset**
- Gabapentin has wide inter-patient variability and nonlinear kinetics; It is also renally cleared
- **Taper gradually over at least 1 week before discontinuing**

Tricyclic Antidepressants

- **Caution: elderly, cerebrovascular disorders, seizure, QT prolongation**
- Onset within 1-2 weeks, trial 6-8 weeks for efficacy
- **Taper before discontinuing**

A top-down view of a grey desk with various items: a stethoscope, a laptop displaying a medical website, a tablet, a pair of glasses, a smartphone, a spiral notebook with a pen, and a coffee cup. A large white pill is also visible on the left side of the desk.

Pharm*Achieve*

PALLIATIVE CARE

Qualifying Exam Preparatory Course

PALLIATIVE/END-OF-LIFE CARE

APPROACH TO THERAPY - COMFORT CARE MEASURES

PAIN TYPE	ADJUVANT USED IN PALLIATIVE CARE
Vertebral compression & bone pain	• Bisphosphonates IV (pamidronate IV, zoledronic acid IV), corticosteroids, NSAIDs
Neuropathic pain	• Gabapentin/pregabalin, TCA, ketamine (NMDA receptor antagonist), cannabinoids, topiramate
Soft tissue pain	• NSAIDs • Corticosteroids • Skeletal muscle relaxants (i.e. cyclobenzaprine)
Visceral Pain	• Corticosteroids • Anti-spasmodics

MONITORING OF CORTICOSTEROIDS

- Monitor for:
 - Neuromuscular and skeletal effects (osteoporosis, fracture, osteonecrosis)
 - Ocular effects (IOP, glaucoma)
 - Cushing's (truncal obesity, facial adipose tissue)
 - GI effects
 - Hyperglycemia
 - Immunosuppression
 - Adrenal insufficiency – cortisol production suppressed (negative feedback)
 - Fluid retention
 - Psychiatric symptoms – apathy, depression
- Tapering recommended after long-term use

PAIN MANAGEMENT PALLIATIVE CARE

- Consider baseline pain medication and incident/breakthrough pain as needed
- Opioids and non-opioid options
- Preference: Long-acting formulations over short-acting oral to increase compliance
- Use of high potency opioids common (hydromorphone/morphine)
 - Subcutaneous – intermittent or continuous
- Ketamine as an adjunct in severe/intractable pain

MANAGING PALLIATIVE SYMPTOMS

Non-pharmacological Measures for nausea

- Limit fatty, spicy and/or salty foods
- Manage environment (smells/noise)
- Relaxation
- Acupressure for chemo-induced acute nausea

GI dysmotility

Metoclopramide

GI obstruction

Haloperidol, metoclopramide, olanzapine

Opioid-induced nausea

Metoclopramide, haloperidol, ondansetron; refractory: olanzapine, risperidone

Vestibular, motion-related nausea

Dimenhydrinate, scopolamine

Chemotherapy-related nausea

Glucocorticoid, NK1R antagonist, 5HT3 antagonist

Increased Intracranial pressure

Corticosteroids, carbamazepine

Emotional/anxiety

Lorazepam

MANAGING PALLIATIVE SYMPTOMS

Agitation and Delirium

Antipsychotics (haloperidol) q4h PRN

Anxiety/Depression

- Treat pain and other reversible physical symptoms first
- **SSRI/SNRI** effective +/- psychosocial counseling
- **TCAs (nortriptyline and desipramine)** for co-analgesic benefit
- **Benzodiazepines** as adjunct to SSRI/SNRI

Respiratory Secretion

- **Scopolamine** PRN (very sedating)
- **Glycopyrrolate** (less sedating) PRN – first choice

Dyspnea

Use opioids with caution for breathlessness

Constipation

- Rule out bowel obstruction
- Regular use of **stimulant** and/or **osmotic laxatives (bisacodyl, sennosides, PEG 3350)**
- Avoid docusate (ineffective)
- Refractory opioid induced constipation
 - Consider **methylnaltrexone** if other options are ineffective
 - Reduce opioid dose, if possible or switch to a less constipating one (fentanyl)

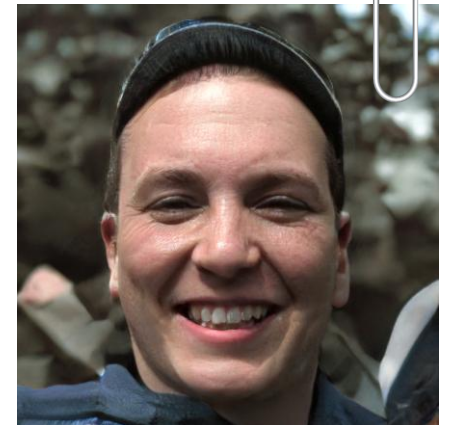
Dry Mouth

- Chewing gum, suck on hard candies or ice chips
- **Saliva substitutes**
- Limit anticholinergic medication
- Oral balance gel

CASE 1

RY is a 50-year-old male that has been a patient at your chronic pain clinic for the past three years. He is finding it hard to manage taking so many pain pills every day and is looking for other options like the fentanyl patch. He presents to your pain clinic for a consultation. His pain is well-controlled with his current opioid regimen. There are no concerns regarding aberrant drug-related behaviors. He has been very compliant with his opioids but is starting to find them inconvenient to take, and they upset his stomach. He has a past medical history of hypertension, dyslipidemia, chronic pain, insomnia, opioid, induced constipation. His pharmacy profile shows he is currently taking ramipril 10 mg PO daily, rosuvastatin 10 mg PO daily, morphine SR 100 mg PO Q12H, morphine IR 5 mg PO BID, lorazepam 1 mg QHS PRN, PEG 3350 17 g daily and senna 8.6 mg QHS. He does not smoke or partake in recreational drugs, while he does drink socially (1/week). His pain intensity is 3/10, and his blood pressure is 130/65 mmHg. He has no allergies to medications.

1. The team has determined that fentanyl transdermal is a good option for RY. What is the most appropriate starting dose?
 - a) 37mcg/hr
 - b) 25mcg/hr
 - c) 50mcg/hr
 - d) 12mcg/hr



FENTANYL CONVERSION

Morphine PO (mg per day)	Fentanyl patch (mcg/h)
60-134	25
135-179	37 (25+12)
180-224	50
225-269	62 (50+12)
270-314	75
315-359	87 (75+12)
360-404	100

UNIDIRECTIONAL

Patch Strength	Single Breakthrough dosing: Morphine IR
12 mcg/hr	5 mg
25 mcg/hr	10 mg
50 mcg/hr	20 mg
75 mcg/hr	25 mg
100 mcg/hr	35 mg



Patch is contraindicated in post-operative, acute pain, and opioid-naïve patients

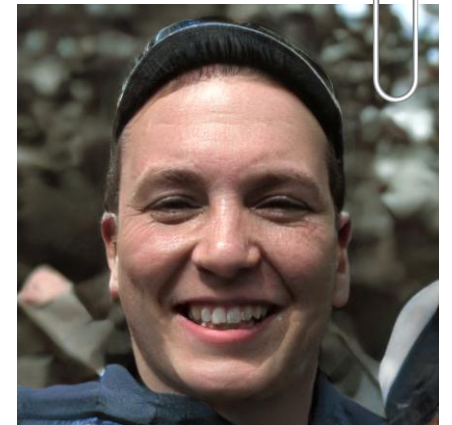


Chart above already accounts for cross-tolerance!

CASE 1

RY is a 50-year-old male that has been a patient at your chronic pain clinic for the past three years. He is finding it hard to manage taking so many pain pills every day and is looking for other options like the fentanyl patch. He presents to your pain clinic for a consultation. His pain is well-controlled with his current opioid regimen. There are no concerns with regard to aberrant drug-related behaviors. He has been very compliant with his opioids but is starting to find them inconvenient to take, and they upset his stomach. He has a past medical history of hypertension, dyslipidemia, chronic pain, insomnia, opioid, induced constipation. His pharmacy profile shows he is currently taking ramipril 10 mg PO daily, rosuvastatin 10 mg PO daily, morphine SR 100 mg PO Q12H, morphine IR 5 mg PO BID, lorazepam 1 mg QHS PRN, PEG 3350 17 g daily and senna 8.6 mg QHS. He does not smoke or partake in recreational drugs, while he does drink socially (1/week). His pain intensity is 3/10, and his blood pressure is 130/65 mmHg. He has no allergies to medications.

2. When should the patient apply the fentanyl transdermal patch in relation to his last oral opioid dose?
 - a) RY should stop morphine SR for 48 hours and then apply the patch
 - b) RY should stop morphine SR for 24 hours, then apply the patch
 - c) RY should stop morphine SR for 12 hours, then apply the patch
 - d) RY should take one dose of morphine SR at the same time that the patch is applied



CASE 1

RY is a 50-year-old male that has been a patient at your chronic pain clinic for the past three years. He is finding it hard to manage taking so many pain pills every day and is looking for other options like the fentanyl patch. He presents to your pain clinic for a consultation. His pain is well-controlled with his current opioid regimen. There are no concerns with regard to aberrant drug-related behaviors. He has been very compliant with his opioids but is starting to find them inconvenient to take, and they upset his stomach. He has a past medical history of hypertension, dyslipidemia, chronic pain, insomnia, opioid, induced constipation. His pharmacy profile shows he is currently taking ramipril 10 mg PO daily, rosuvastatin 10 mg PO daily, morphine SR 100 mg PO Q12H, morphine IR 5 mg PO BID, lorazepam 1 mg QHS PRN, PEG 3350 17 g daily and senna 8.6 mg QHS. He does not smoke or partake in recreational drugs, while he does drink socially (1/week). His pain intensity is 3/10, and his blood pressure is 130/65 mmHg. He has no allergies to medications.

3. When can the dose of fentanyl transdermal be increased and titrated up?

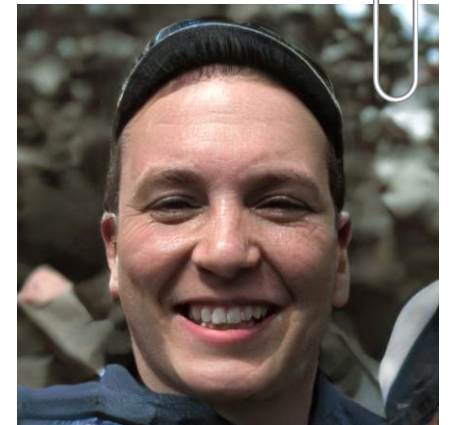
- a) 7 days after the initial patch
- b) 3 days after the initial patch
- c) 2 days after the initial patch
- d) 1 day after the initial patch



CASE 1

RY is a 50-year-old male that has been a patient at your chronic pain clinic for the past three years. He is finding it hard to manage taking so many pain pills every day and is looking for other options like the fentanyl patch. He presents to your pain clinic for a consultation. His pain is well-controlled with his current opioid regimen. There are no concerns with regard to aberrant drug-related behaviors. He has been very compliant with his opioids but is starting to find them inconvenient to take, and they upset his stomach. He has a past medical history of hypertension, dyslipidemia, chronic pain, insomnia, opioid, induced constipation. His pharmacy profile shows he is currently taking ramipril 10 mg PO daily, rosuvastatin 10 mg PO daily, morphine SR 100 mg PO Q12H, morphine IR 5 mg PO BID, lorazepam 1 mg QHS PRN, PEG 3350 17 g daily and senna 8.6 mg QHS. He does not smoke or partake in recreational drugs, while he does drink socially (1/week). His pain intensity is 3/10, and his blood pressure is 130/65 mmHg. He has no allergies to medications.

4. RY states he is experiencing constipation even after switching to fentanyl. He also reports the PEG and senna are no longer helping. Which of the following is an option for RY's refractory OIC?
- a) Increase PEG 3350 to twice daily
 - b) Increase Senna to twice daily
 - c) Trial mineral oil
 - d) Trial methylnaltrexone



REFERENCES

1. Rosenquist EW. Definition and pathogenesis of chronic pain. UpToDate. <http://www.uptodate.com/definition-and-pathogenesis-of-chronic-pain>. Updated January 9, 2015.
Rosenquist EW. Overview of the treatment of chronic pain. UpToDate. <http://www.uptodate.com/contents/overview-of-the-treatment-of-chronic-pain>. Updated February 3, 2017.
2. Wiechman S. Burn pain: Principles of pharmacologic and nonpharmacologic management. UpToDate. <https://www.uptodate.com/contents/burn-pain-principles-of-pharmacologic-and-nonpharmacologic-management>. Updated February 2017.
3. Rosenquist E. Evaluation of chronic pain in adults. UpToDate. <https://www.uptodate.com/contents/evaluation-of-chronic-pain-in-adults>. Updated February 2017.
4. Neuropathic Pain. In: RxTx. Ottawa, ON: Canadian Pharmacists Association. [Updated October 2014].
5. <https://myrxtx.ca/new/documents/CHAPTER/en/c0012>.
6. Tricyclic antidepressants (CPhA monograph). In: RxTx. Ottawa, ON: Canadian Pharmacists Association. [Updated December 2016].
<https://myrxtx.ca/new/documents/MONOGRAPH/en/CyclicAntidepressantsCPhA>.
7. Gupta A, Rosenquist R. Use of opioids in the management of chronic non-cancer pain. UpToDate. <https://www.uptodate.com/contents/use-of-opioids-in-the-management-of-chronic-non-cancer-pain>. Updated February 2017.
8. Canadian Guideline for Opioid Use for Pain — Appendix B-8: Opioid Conversion and Brand Availability in Canada.
http://nationalpaincentre.mcmaster.ca/opioid/cgop_b_app_b08.html.
9. Non-Pharmacological Methods for Managing Chronic Pain: Physical Methods. The Canada Agency for Drugs and Technologies in Health (CADTH). September 2020.
https://www.cadth.ca/sites/default/files/tools/Chronic%20Pain/Clinician%20Resources/National%20versions/clinician_summary_physical_FINAL.pdf
10. Mu A, Weinberg E, Moulin DE, Clarke H. Pharmacologic management of chronic neuropathic pain. Review of the Canadian Pain Society consensus statement. *Can Fam Physician*. 63:844-852. November 2017.
11. Bhat, C, Rosenberg, H, James, D. Topical nonsteroidal anti-inflammatory drugs. September 2023. CMAJ. <https://www.cmaj.ca/content/cmaj/195/36/E1231.full.pdf>
12. Furlan AD, Reardon R, Weppeler C. Opioids for chronic noncancer pain: a new Canadian practice guideline. CMAJ 2010. DOI:10.1503/cmaj.100187.
<https://www.cmaj.ca/content/cmaj/suppl/2010/05/03/cmaj.100187.DC1/opioids-weppeler-1-at.pdf>.
13. BGP Pharma ULC. LYRICA® (pregabalin) capsules, 25, 50, 75, 150, 225, 300 mg, oral. Product Monograph. BGP Pharma ULC; 2023. Submission Control Number: 274588.
Available at: https://pdf.hres.ca/dpd_pm/00070660.PDF. Accessed February 26, 2025.
14. Teva Canada Limited. DULOXETINE DR (duloxetine as duloxetine hydrochloride) 30 mg and 60 mg delayed-release capsules. Product Monograph. Teva Canada Limited; October 7, 2016. Control No.: 198591. Available at: https://pdf.hres.ca/dpd_pm/00036663.PDF. Accessed February 26, 2025.
15. Fraser Health. Principles of Opioid Management. https://www.fraserhealth.ca/-/media/Project/FraserHealth/FraserHealth/Health-Professionals/Professionals-Resources/Hospice-palliative-care/Sections-PDFs-for-FH-Aug31/9524-25-FH---Sym_Guide-PrinciplesOfOpioidMgmt.pdf?sc_lang=en.

CHANGE LOG

July 2025

- Minor formatting, spelling, and grammar changes throughout
- Slides 4-7: Reordered for flow, split causes and definitions for clarity
- Slide 6: Expanded on pain causes
- Slides 8-9: Moved topical/localized pharmacotherapy options to their own slide from non-pharmacologic options
- Slide 14: Added genetic polymorphisms of CYP2D6 affect efficacy of codeine therapy (removed from earlier slide)
- Slide 14: Edited language regarding oxycodone misuse
- Slide 15: Edited language regarding heroin use
- Slide 17: Added note to set realistic expectations
- Slide 21-22: Split into separate slides
- Slide 22: Added note incomplete cross-tolerance is not accounted for when going IV to PO for same molecule
- Slide 23 & 38: Edited morphine PO to fentanyl patch
- Slide 31: Added section title page for clear break in content
- Old slide 38 removed due to duplication

September 2025

- Slide 21: Removed parenteral column. Removed tramadol equivalence dose since can not be reliably established

November 2025

- Minor phrasing and formatting changes throughout
- Slide 17: Minor formatting change for added clarity on when to end titration
- Slides 25-26: Updated to 4 choices to align with MCQ exam format
- Slide 40: Updated answer choices

April 2026

- Slide 23: Updated 75mcg and 100mcg breakthrough doses
- Slide 25: New practice question
- Slide 42: New question
- Slide 43: Updated references